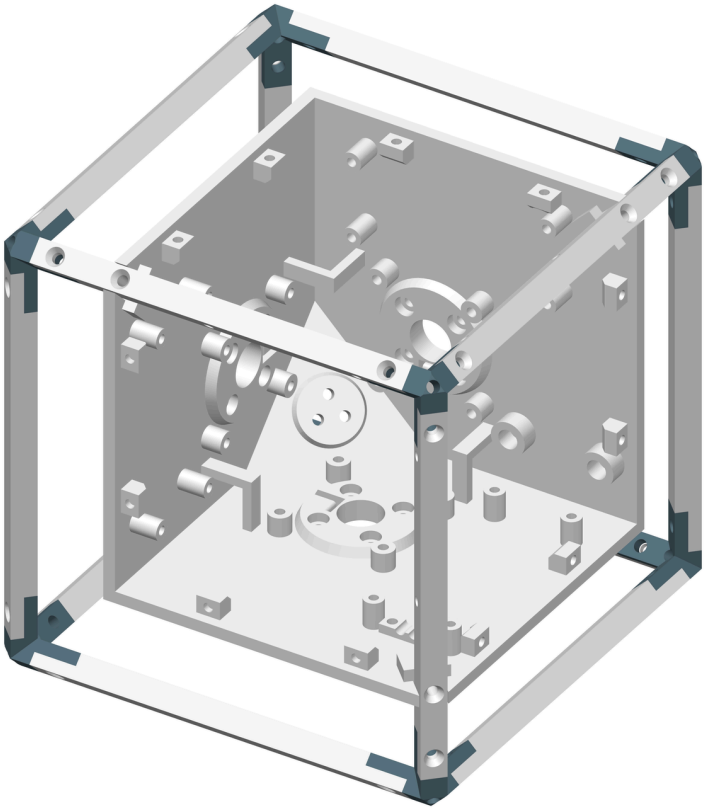
Heat-set inserts

5 steps, about 1 minute per insert.



Designed by Athanasia Nikolova

All data come from tests and many real builds, not only from theory. v1.2 · November 2026

Teacher only



One thing per page

One action, one big picture. Short sentences, much white space.



Color is never the only clue

Every color also has a number, a letter or a name next to it. The part colors stay distinct with color-vision deficiency (Okabe-Ito palette).



Clear print

Instructions at 14 pt, everything else at least 12 pt, in Atkinson Hyperlegible. The text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).



For a screen reader

The PDF is tagged (PDF/UA-1): headings, lists and reading order are marked, and every picture has a text alternative that is read aloud.



Print it right

In color, at 100 % - not 'fit to page'.



Pace

One page, then the action, then the next page. For a student who tires quickly: one lesson per session. The check at the end of each lesson is a good place to stop.

Safety first



Grown-up nearby

Work with your teacher or mentor in the room.



Hot!

The tip is 220 °C. Put it back in the stand every time.



Fresh air

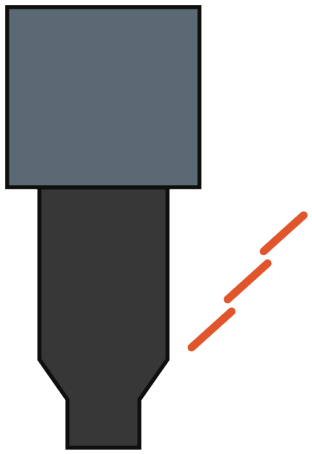
Melted plastic smells. Open a window.



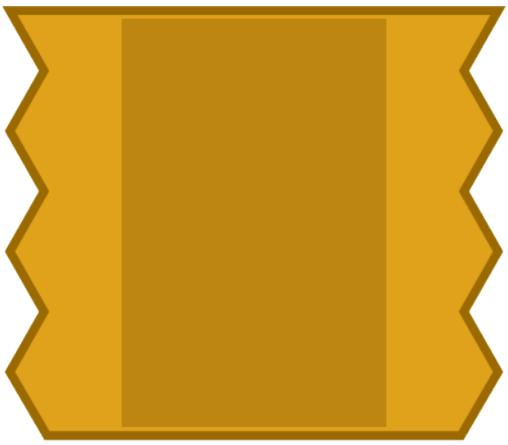
Unplug

When you stop, switch the iron off and let it cool.

You need



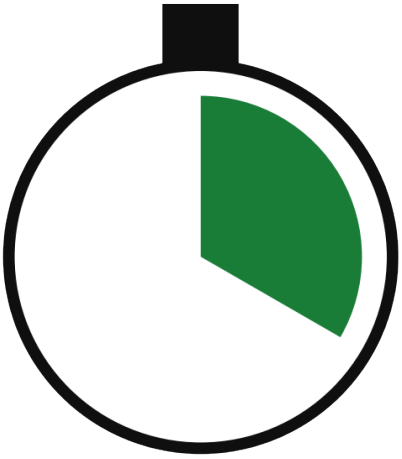
Soldering iron



Brass inserts



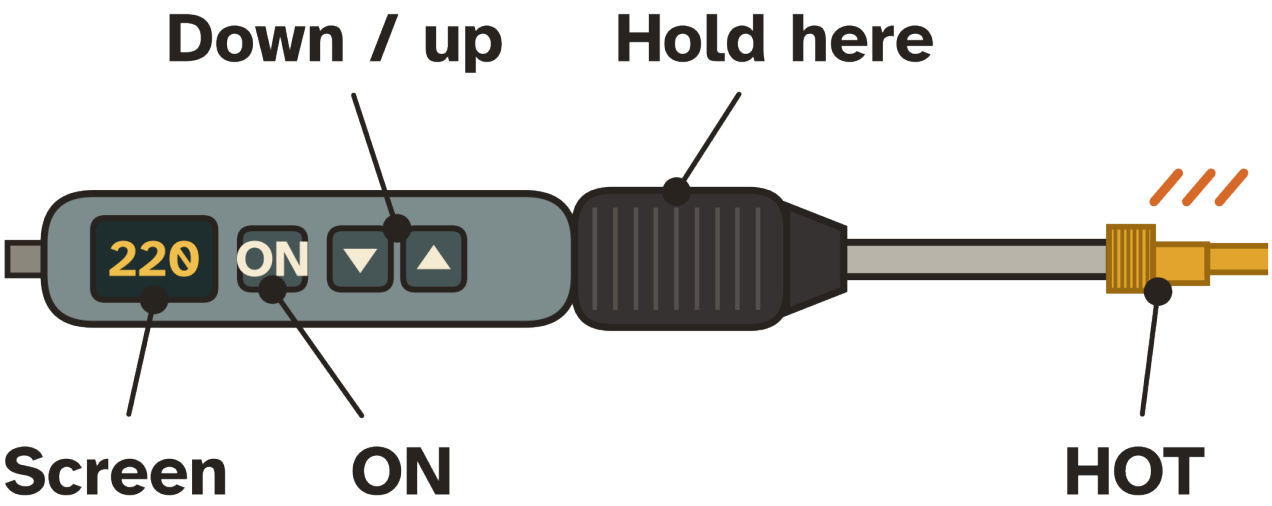
Flat metal block



10 seconds

Your iron

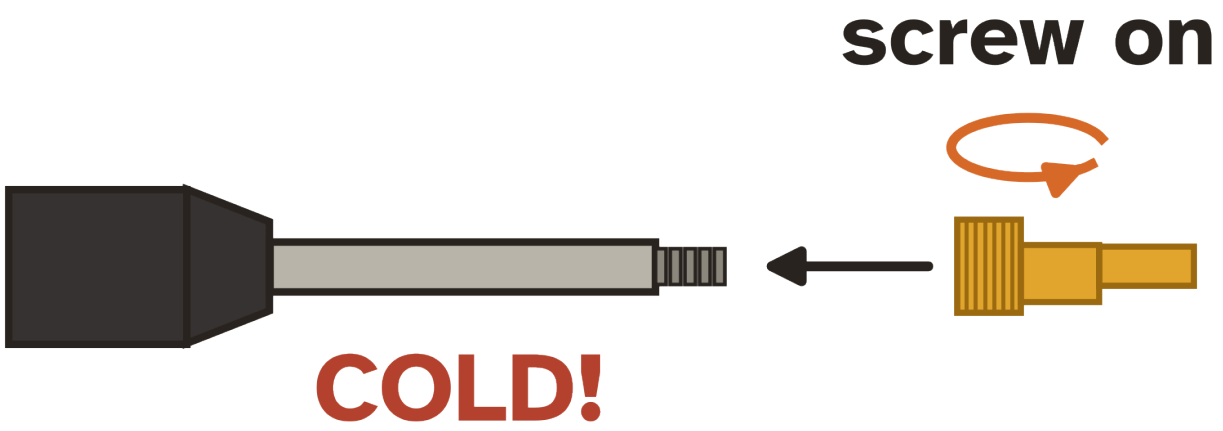
USB-C soldering iron



100–450 °C · 100 W · USB-C · Hot in seconds

- Hold the black grip. Never the metal.

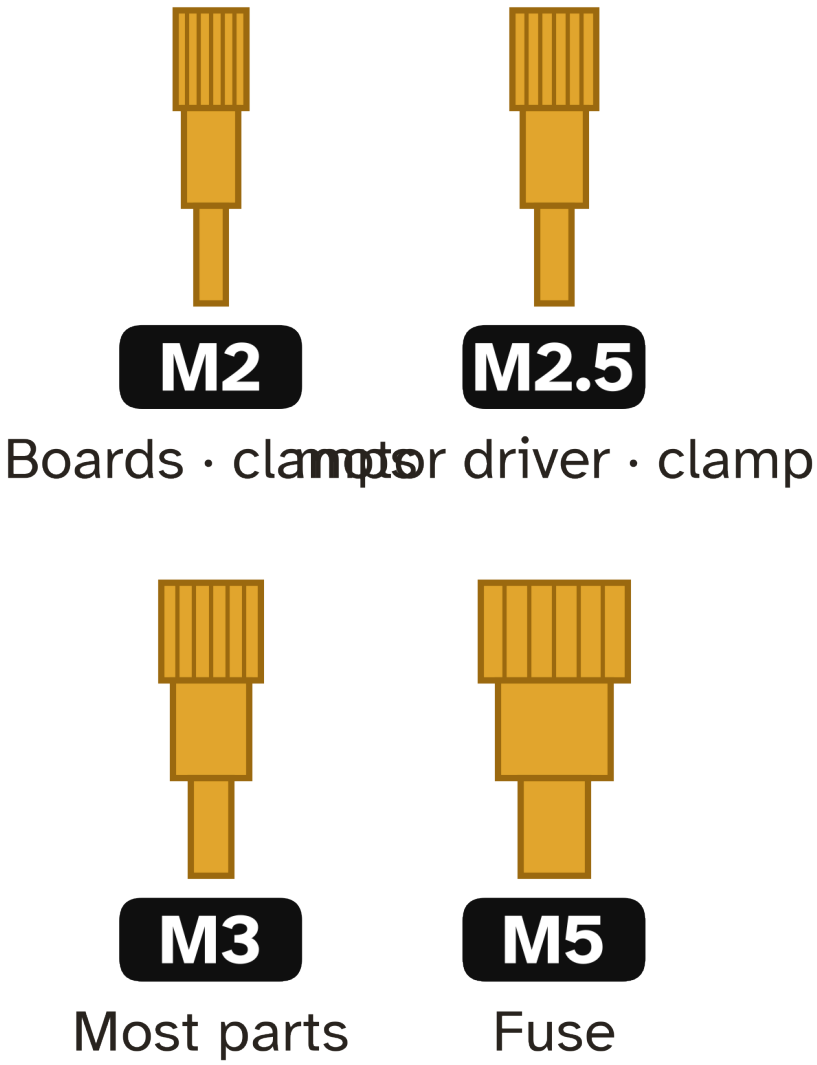
Put on the insert tip



- Iron OFF. Tip cold.
- Screw the brass tip on by hand.

Which tip?

Same size as the insert.



- Change the tip only when it is cold.

Know your inserts

Each kind has its own color, and its name next to it.



M3 SHORT 4 mm · ×91



M3 SHORT 3 mm · ×3



M3 NORMAL 5.7 mm · ×14



M2.5 4 mm · ×14



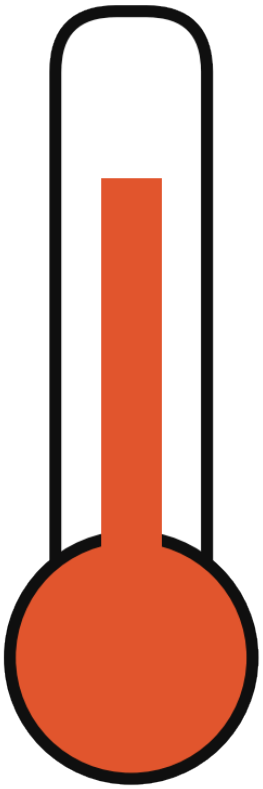
M2 3 mm · ×12



M5 5.8 mm · ×2

Drawn at the same scale. All M3 use the same hole: Ø4.0.

Set the iron

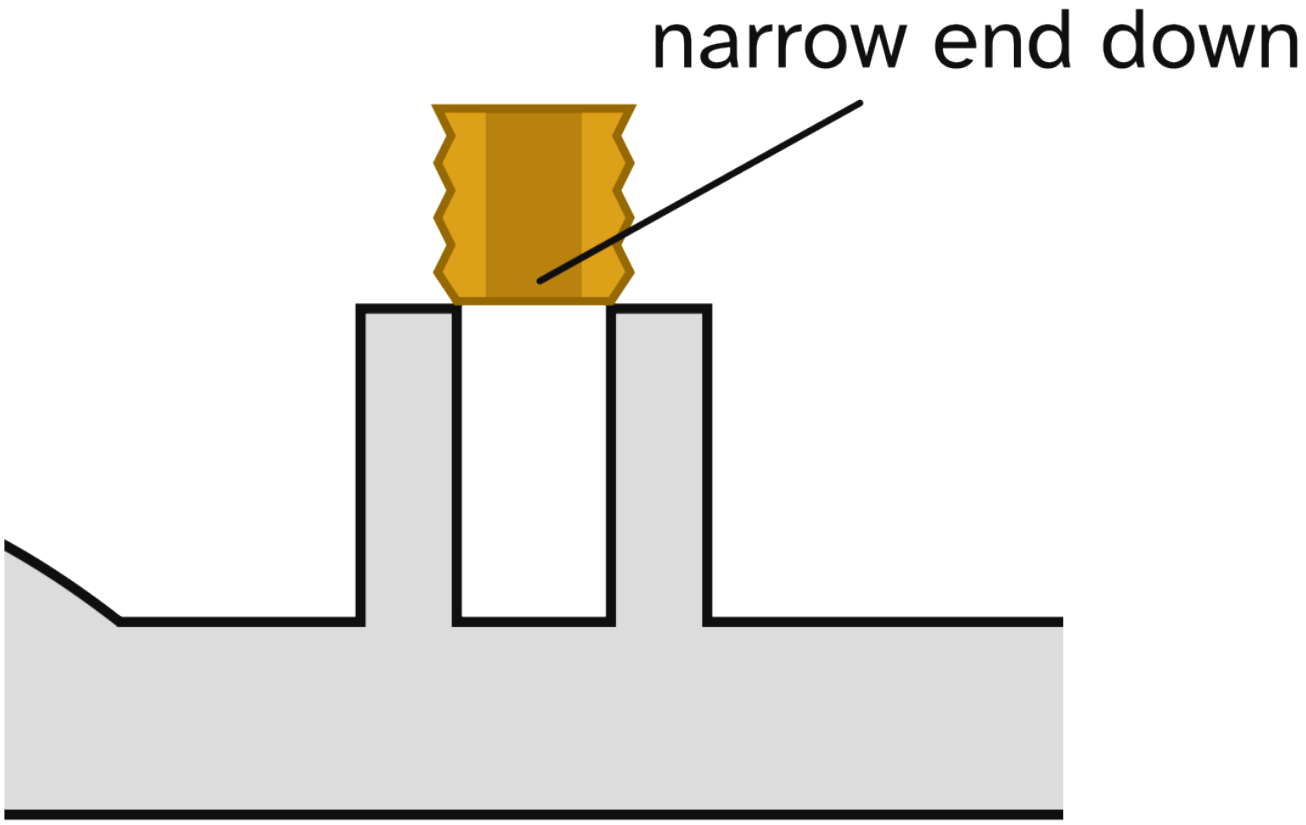


220 °C · PETG-CF

- Up / down to 220.
- Wait until it is hot.

1 • Place

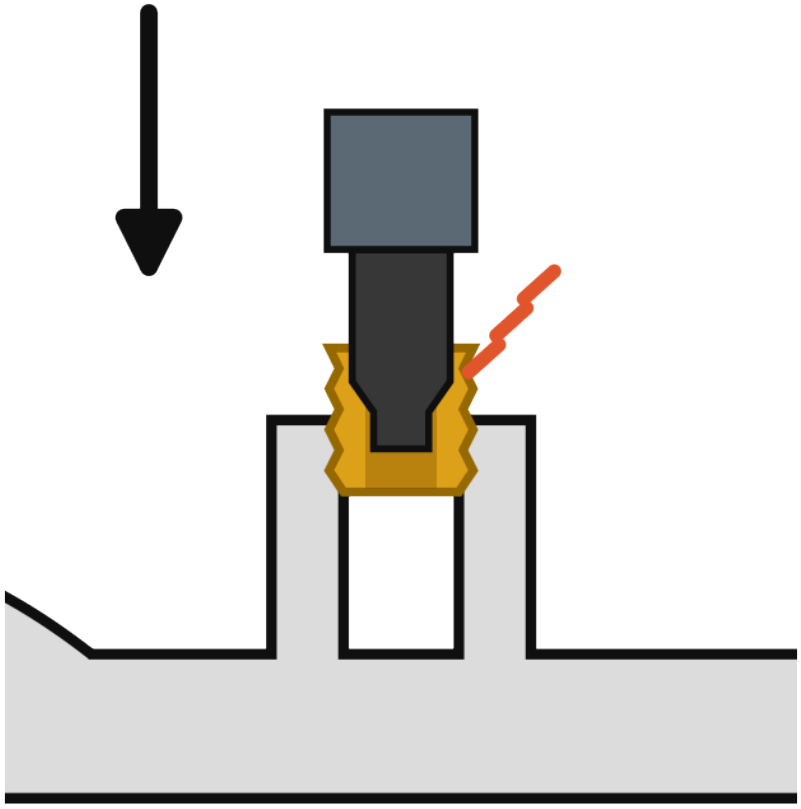
Cut through a real motor driver standoff (box, M2.5)



- Narrow end down.
- Straight on the hole.

2 · Press

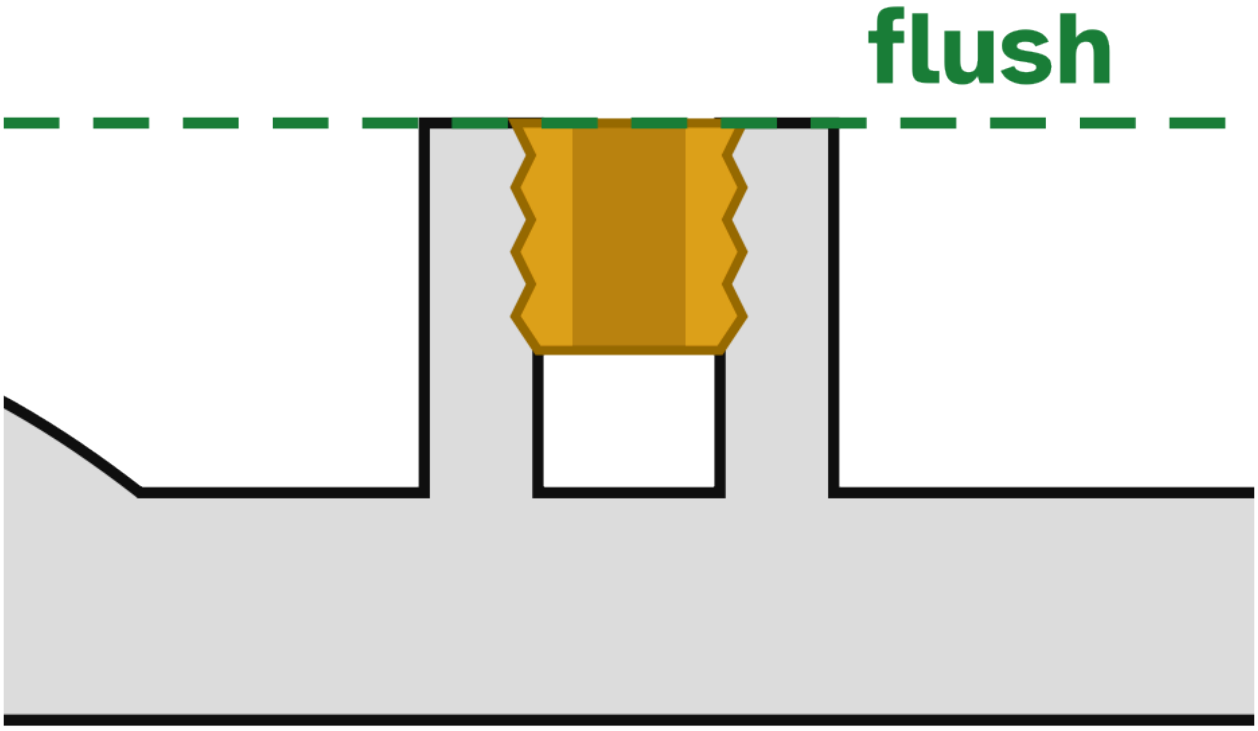
Cut through a real motor driver standoff (box, M2.5)



- Slow and straight.
- Let the heat do the work.

3 · Stop when level

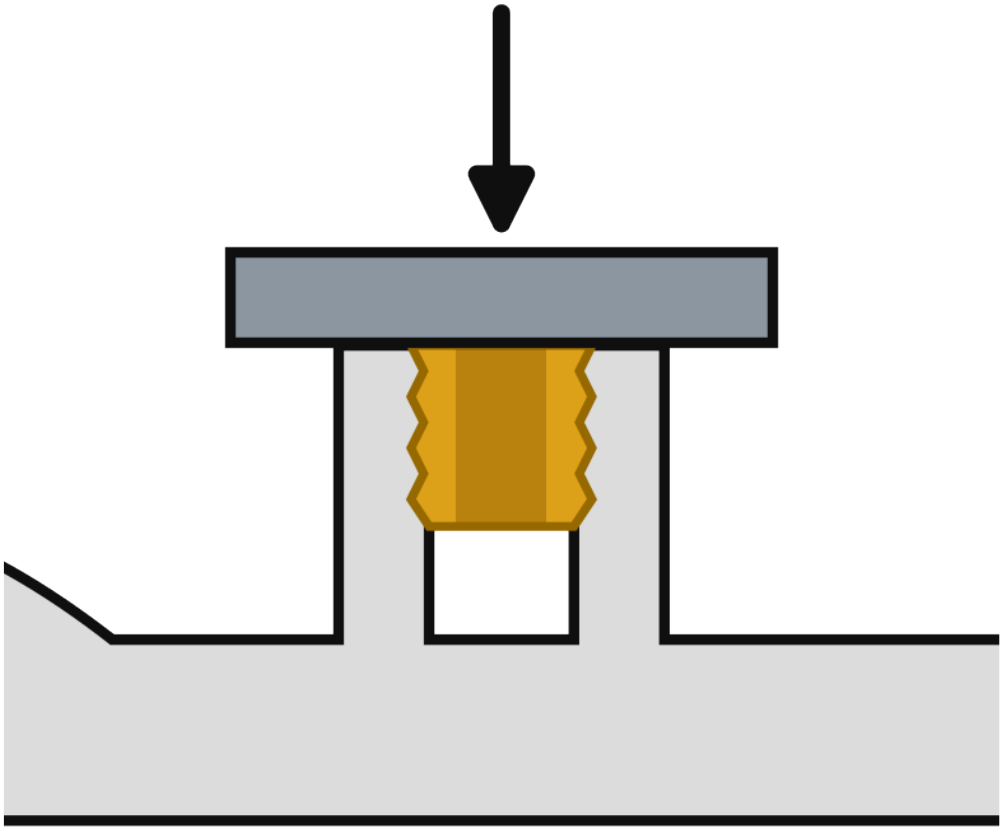
Cut through a real motor driver standoff (box, M2.5)



- Top of the insert = top of the part.

4 · Hold flat

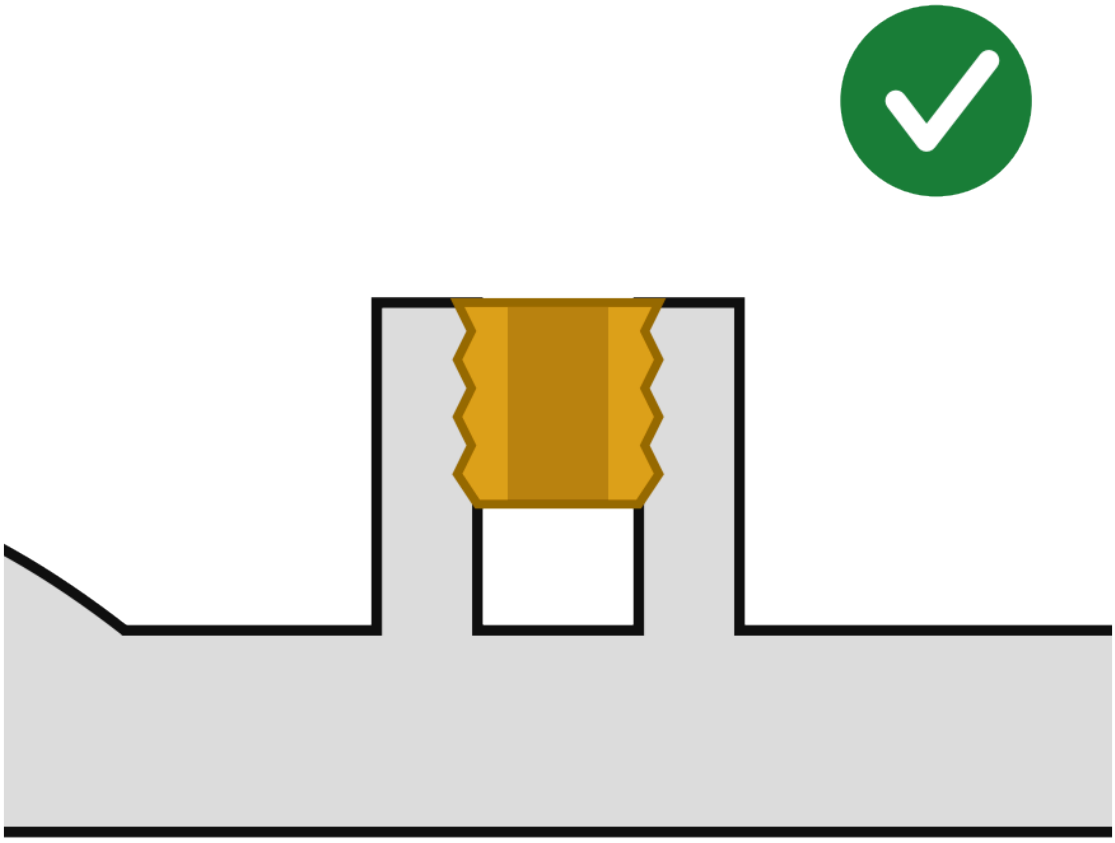
Cut through a real motor driver standoff (box, M2.5)



- Press the metal block on top.
- Count to 10.

5 · Check

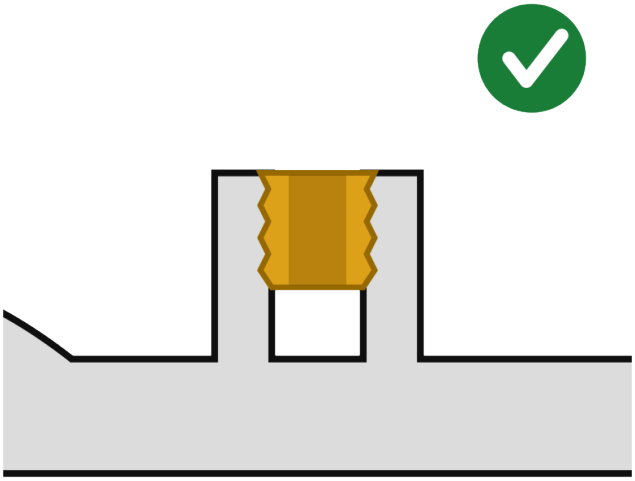
Cut through a real motor driver standoff (box, M2.5)



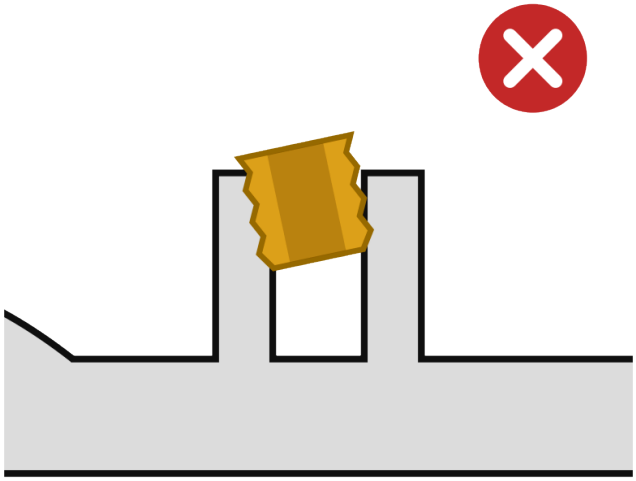
- Straight? Level?
- Done. Next one.

Good or not?

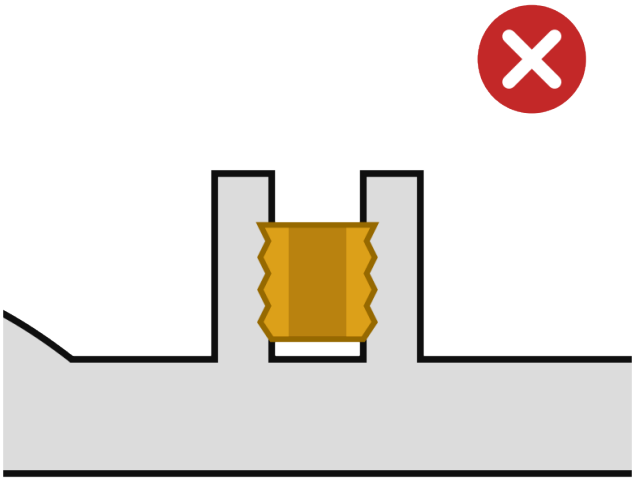
Cut through a real motor driver standoff (box, M2.5)



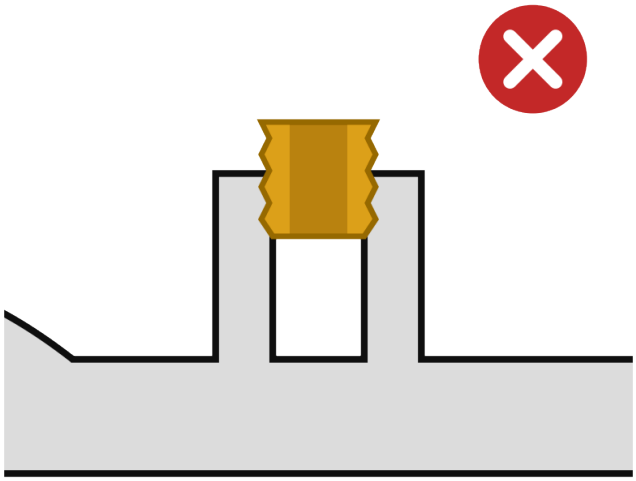
Good



Tilted - reheat, straighten



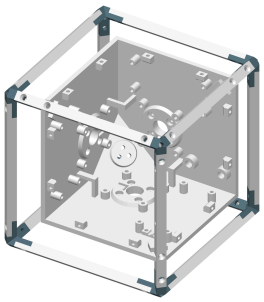
Too deep - the screw must still reach



Sticks out - reheat, press flat

Where they go

Tick each place when it is done.

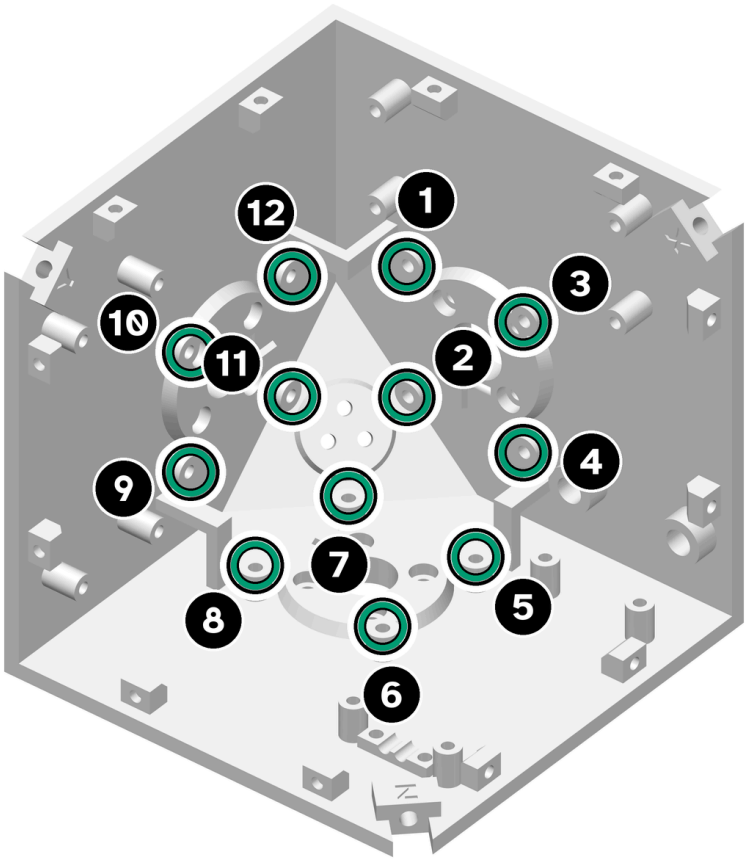


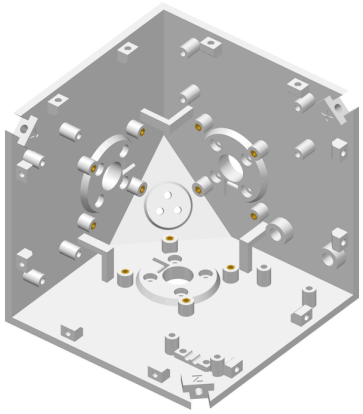
	place	kind	page
☐	Box inside: motor driver	×12	17
☐	Box inside: boards	×12	18
☐	Box inside: fuse	×2	19
☐	Box inside: clamp	×2	20
☐	Box outside: corner 0	×3	21
☐	Box outside: ties	×3	22
☐	Tube: cone end	×3	23
☐	Tube: IMU pocket	×4	24
☐	Tube: table top	×4	25
☐	Basin: top plate	×3	26
☐	Corners: arms	×3 on 8 parts	27
☐	Corners: tip	×1 on 8 parts	28
☐	2 corners: inside	×1 on 2 parts	29
☐	Navels: legs	×6 on 8 parts	30
☐	Lower arcs: middle	×1 on 6 parts	31

Box · inside

Motor faces: 4 standoffs on each

- **M2.5 4 mm** ×12





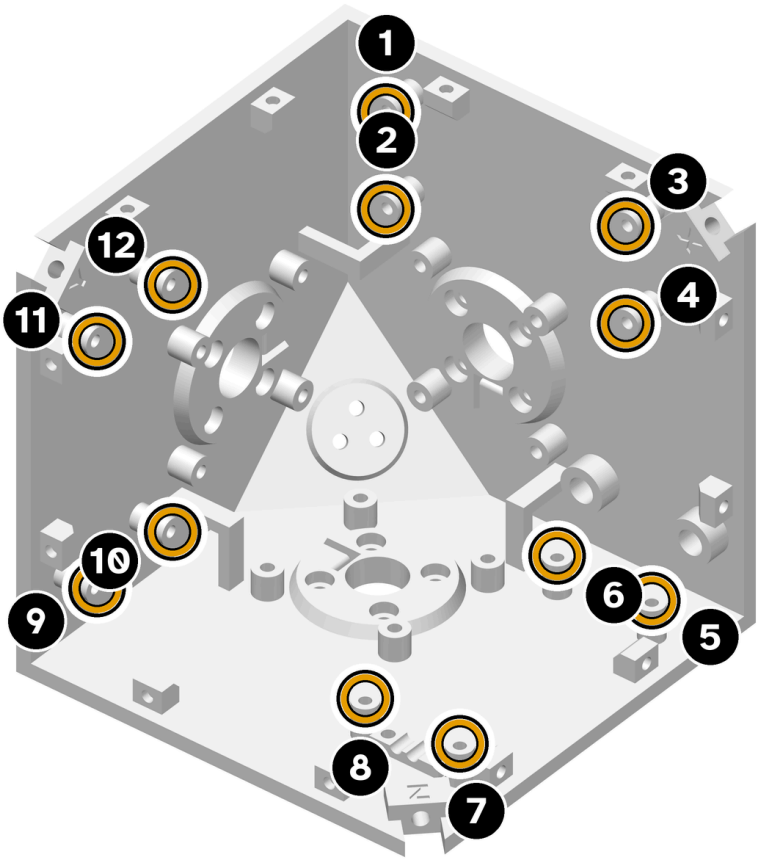
Done: like this

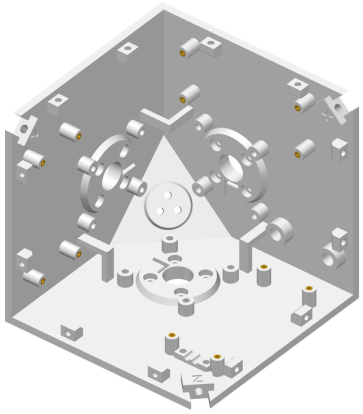
- Later a screw goes in here: motor driver standoffs.

Box · inside

Board standoffs: 4 under each board

- **M2 3 mm** ×12





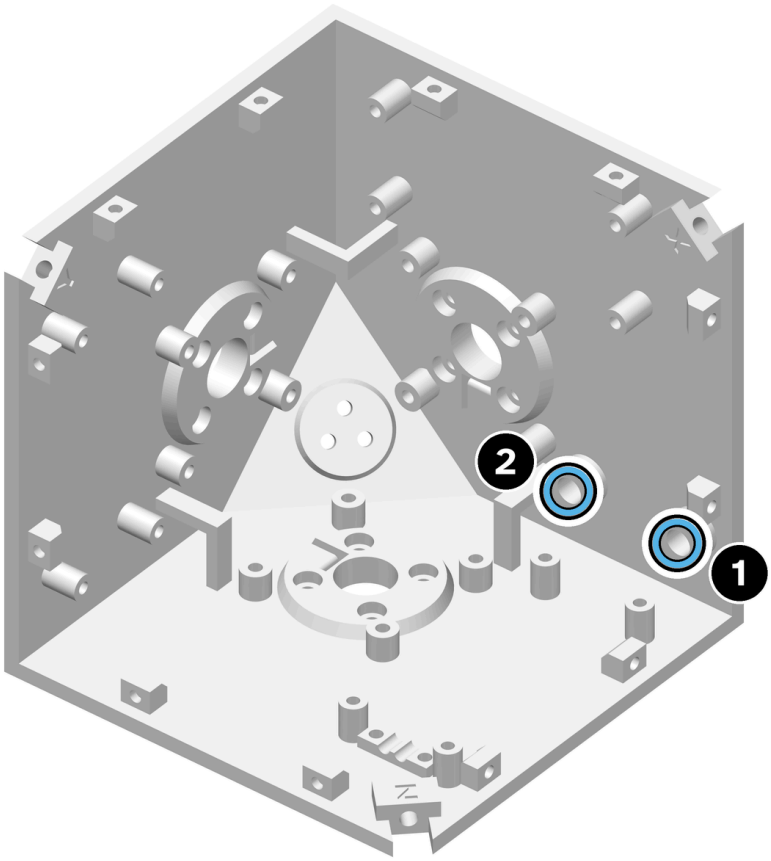
Done: like this

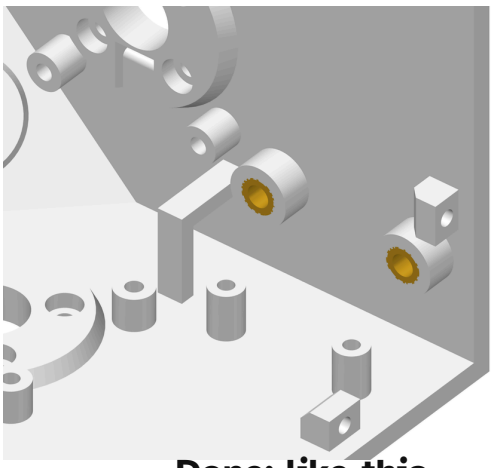
- Later a screw goes in here: board standoffs.

Box · inside

The 2 fuse bosses, face x

- **M5 5.8 mm** ×2





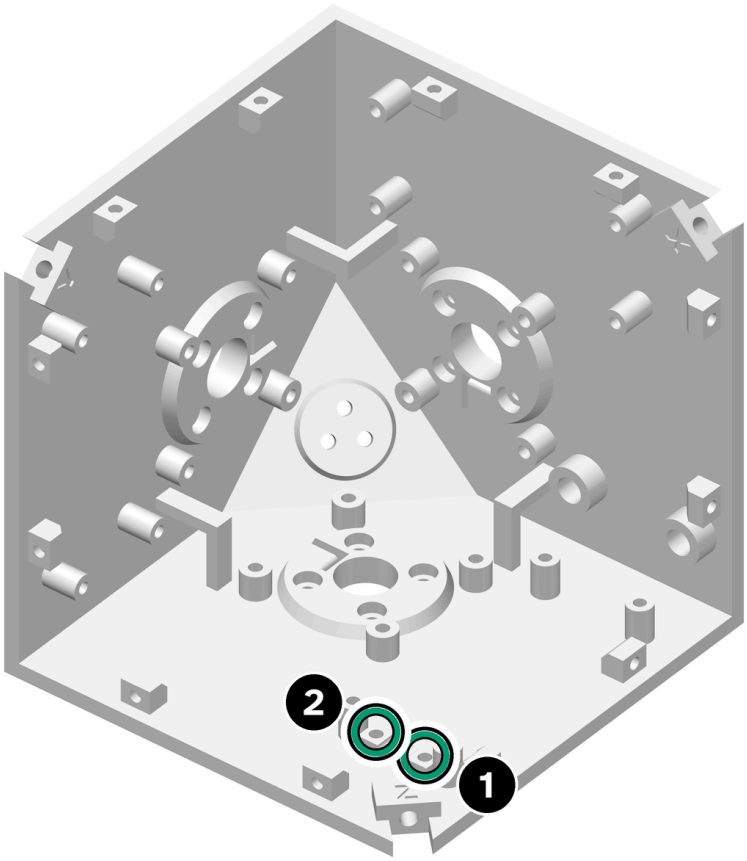
Done: like this

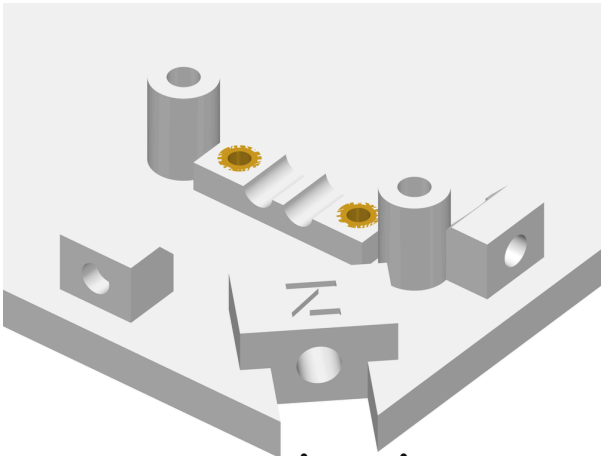
- Later a screw goes in here: MIDI fuse terminals.

Box · inside

Cable clamp base, face z

- **M2.5 4 mm** ×2





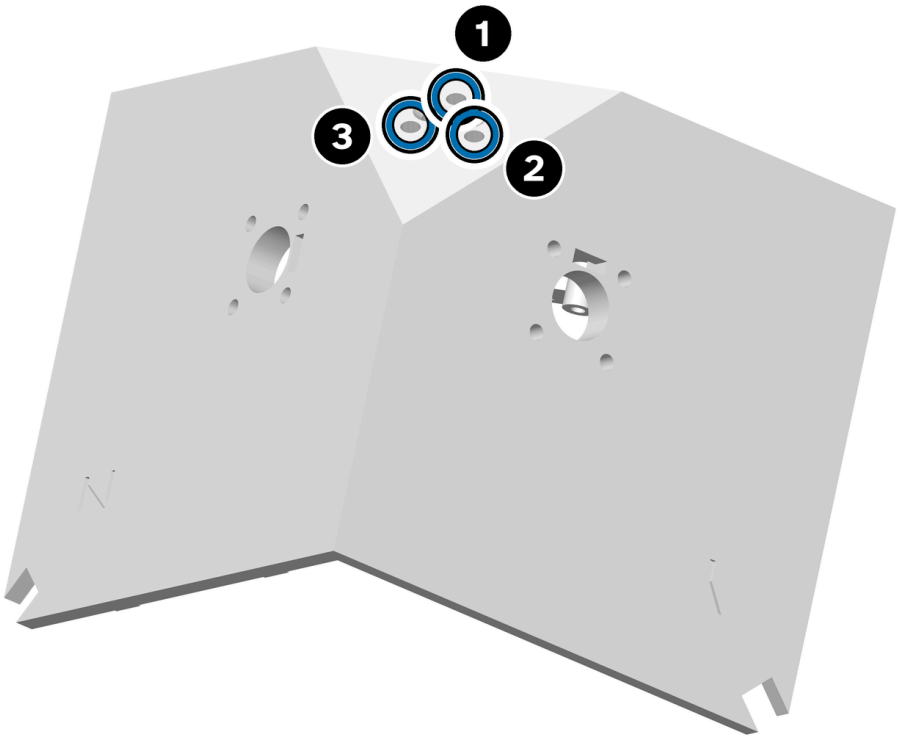
Done: like this

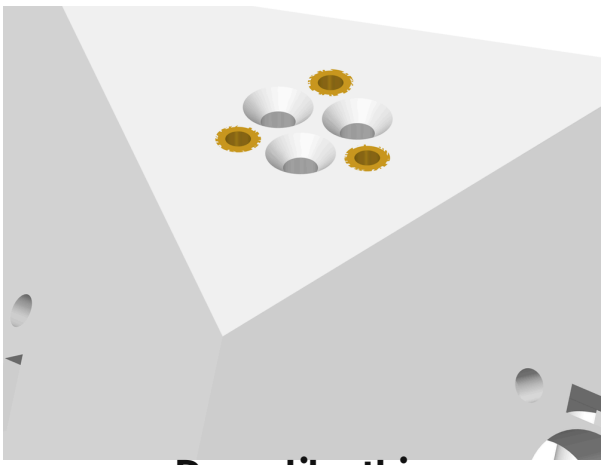
- Later a screw goes in here: cable clamp bar.

Box · outside

Turn the box over: corner 0 up

- **M3 SHORT 4 mm** ×3





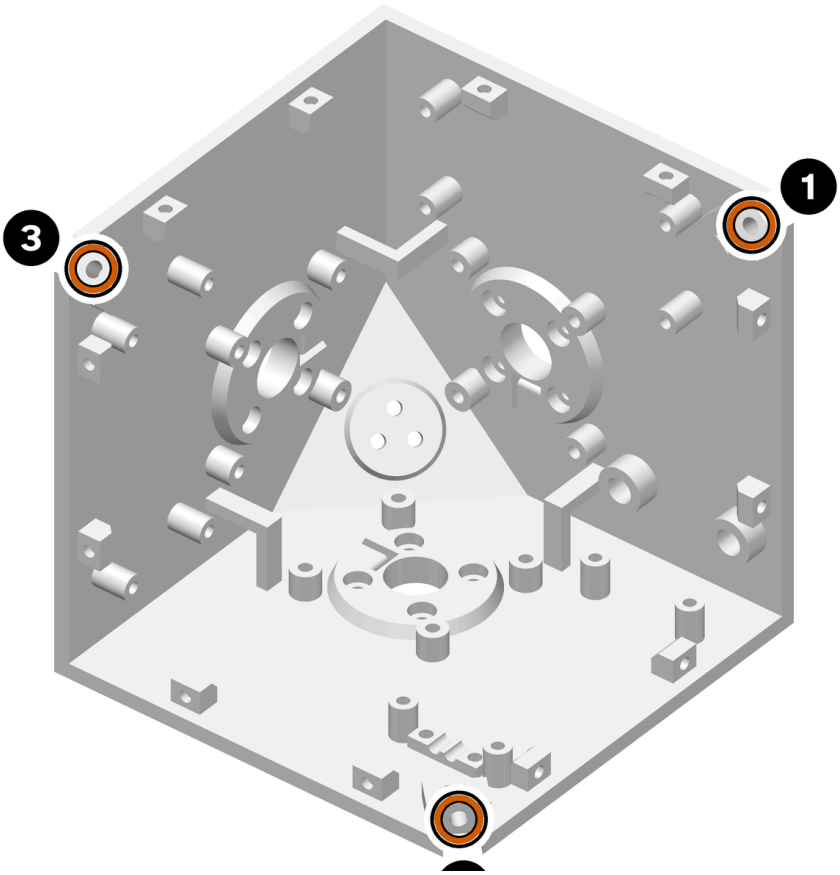
Done: like this

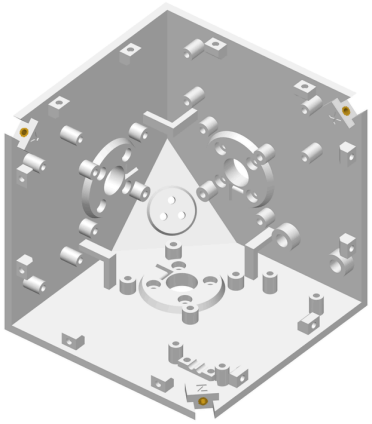
- Later a screw goes in here: box standoff flange.
- Here goes the box standoff flange.

Box · outside

The 3 cage ties, on the edges

- **M3 NORMAL 5.7 mm** ×3





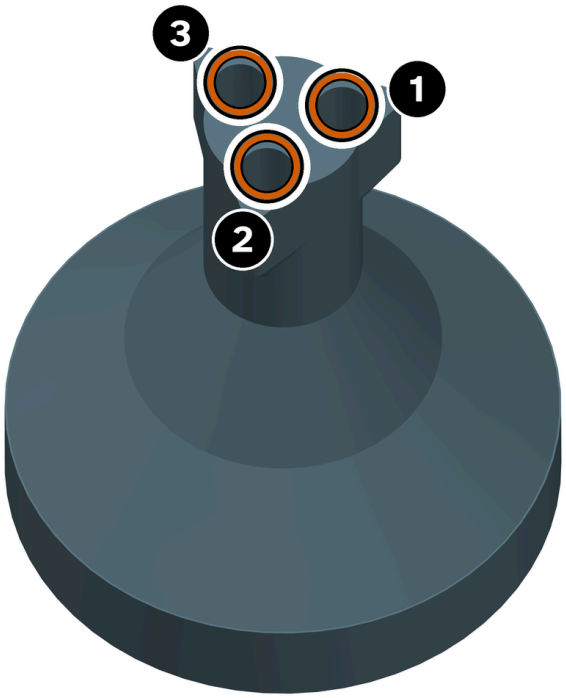
Done: like this

- Later a screw goes in here: cage to Cubi.

Battery tube

Cone end up - as it comes off the printer

- **M3 NORMAL 5.7 mm** ×3





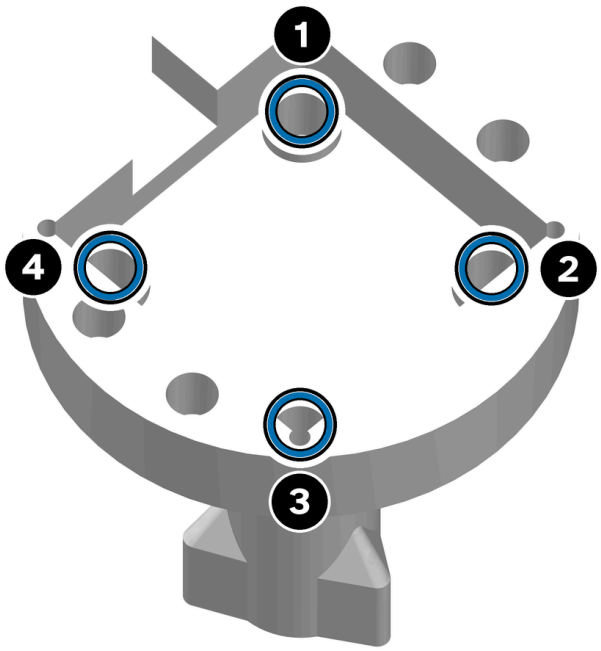
Done: like this

- Later a screw goes in here: from outside the box.

Battery tube

Turn it over: the table up - the 4 in the pocket

- **M3 SHORT 4 mm** ×4





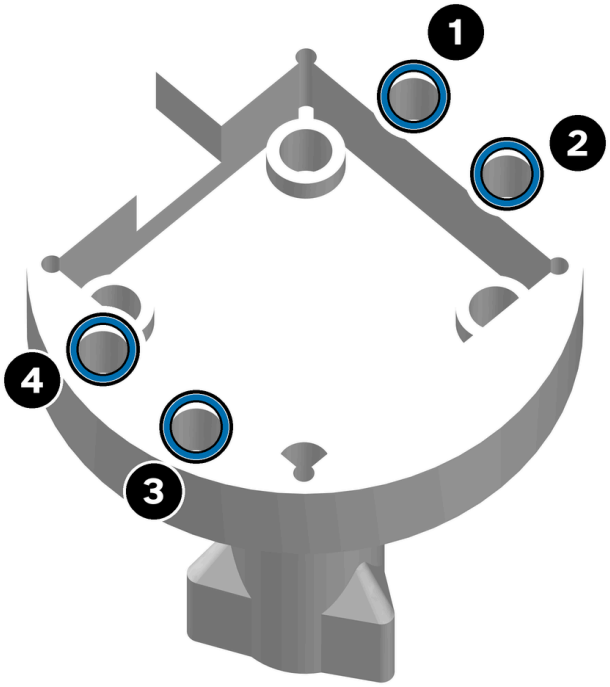
Done: like this

- Later a screw goes in here: IMU screws.
- The IMU board sits in this pocket, on its 4 bosses.

Battery tube

The table up - the 4 on the table top

- **M3 SHORT 4 mm** ×4





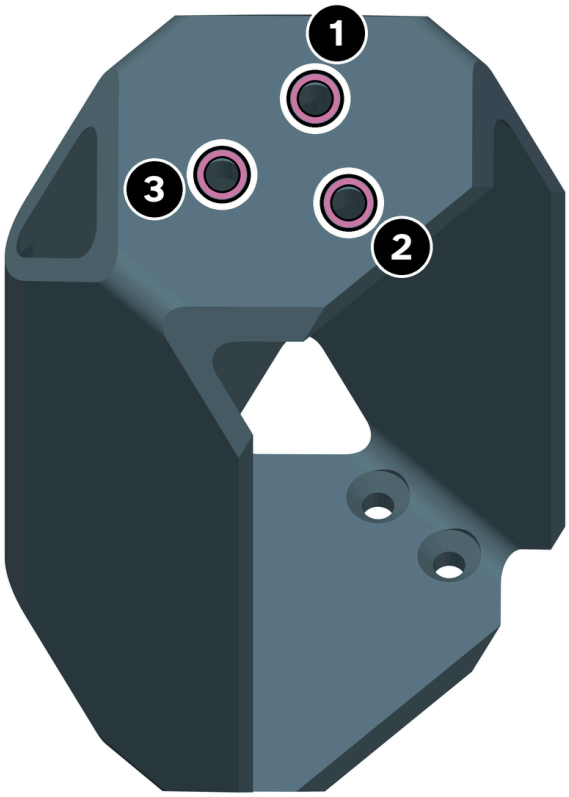
Done: like this

- Later a screw goes in here: basin screws.
- The basin is screwed here, over the IMU.

Battery basin

Top plate, from the outside

- **M3 SHORT 3 mm** ×3





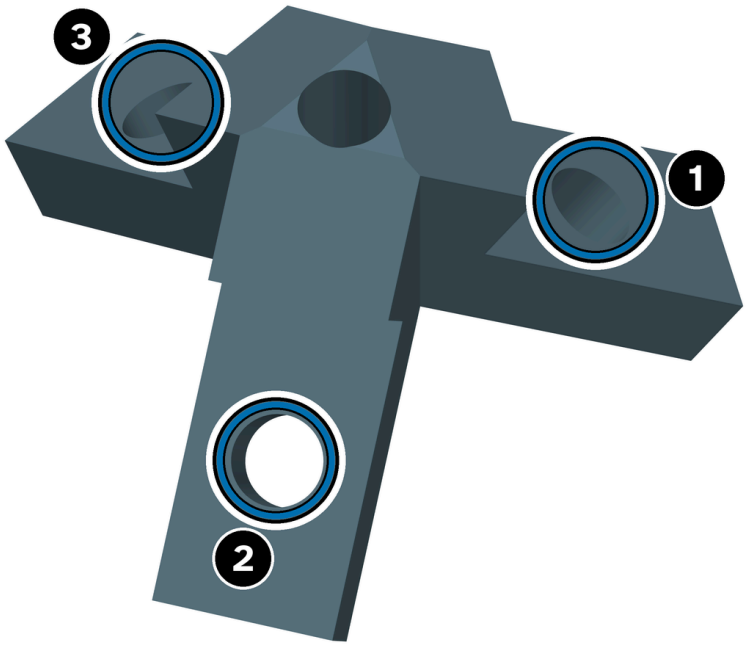
Done: like this

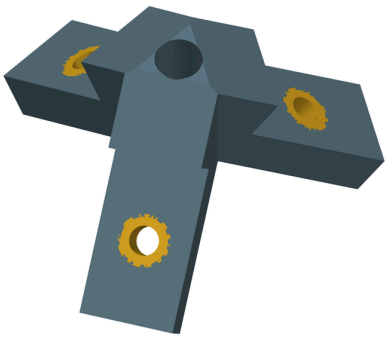
- Later a screw goes in here: hub standoff flange.

Cage corner · all 8

The 3 arm ends, from the outside

- **M3 SHORT 4 mm** ×3 · on each part: 8 parts = 24 inserts





Done: like this

- Later a screw goes in here: cage beams.

Cage corner · all 8

The tip, from the outside

- **M3 NORMAL 5.7 mm** ×1 · on each part: 8 parts = 8 inserts





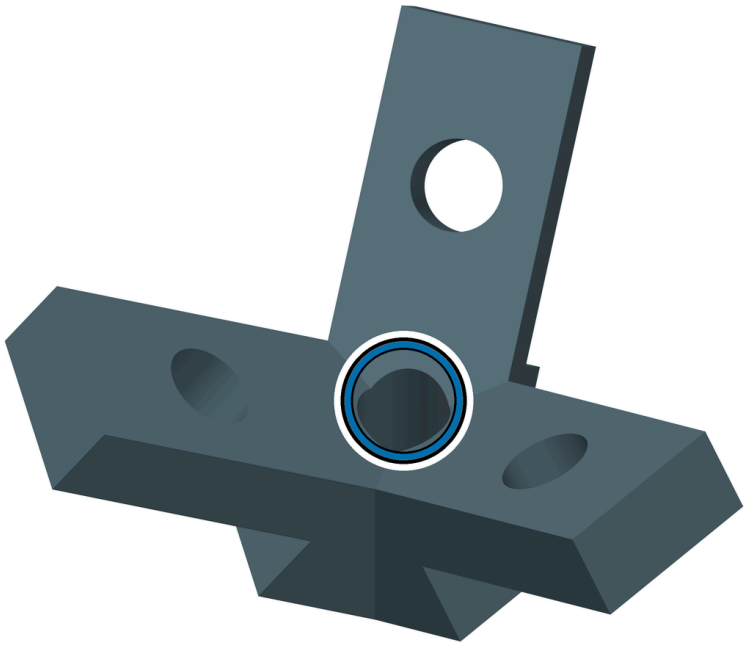
Done: like this

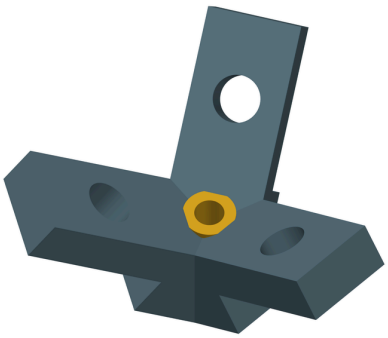
- Later a screw goes in here: navel.
- 2 corners get one more insert, inside - next page.

Cage corner · only 2

The tip, from the INSIDE

- **M3 SHORT 4 mm** ×1 · on each part: 2 parts = 2 inserts





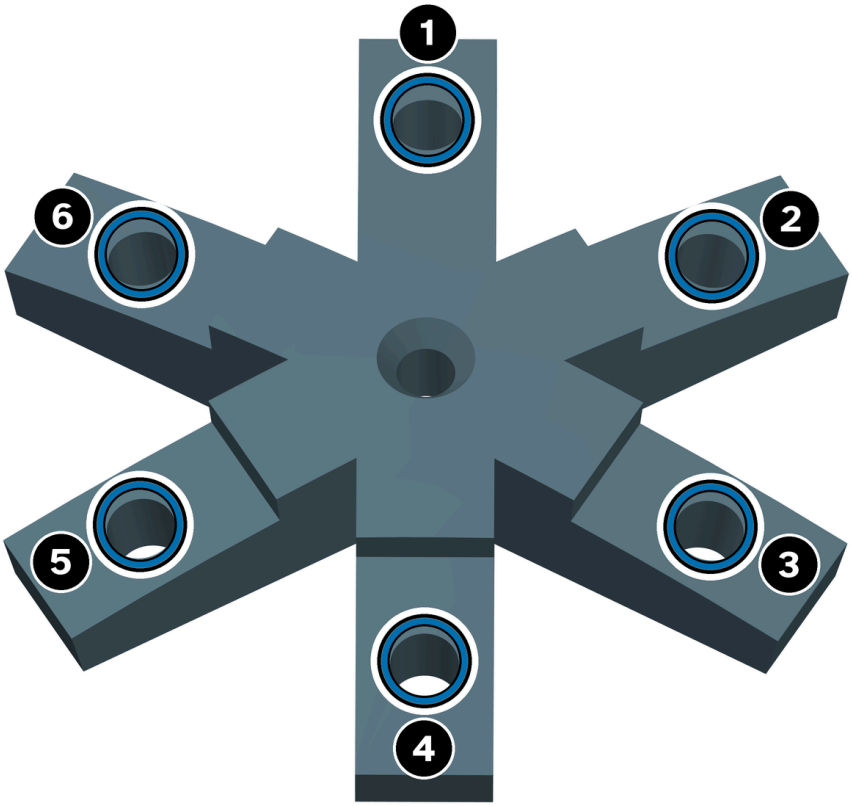
Done: like this

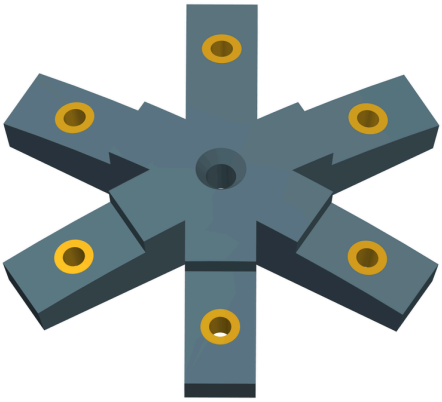
- Later a screw goes in here: box standoff / hub standoff.
- Pick 2 corners, mark them with a pencil: corner 0 and corner (1,1,1). Same hole as the tip insert.

Navel · all 8

The 6 legs, from the top

- **M3 SHORT 4 mm** ×6 · on each part: 8 parts = 48 inserts





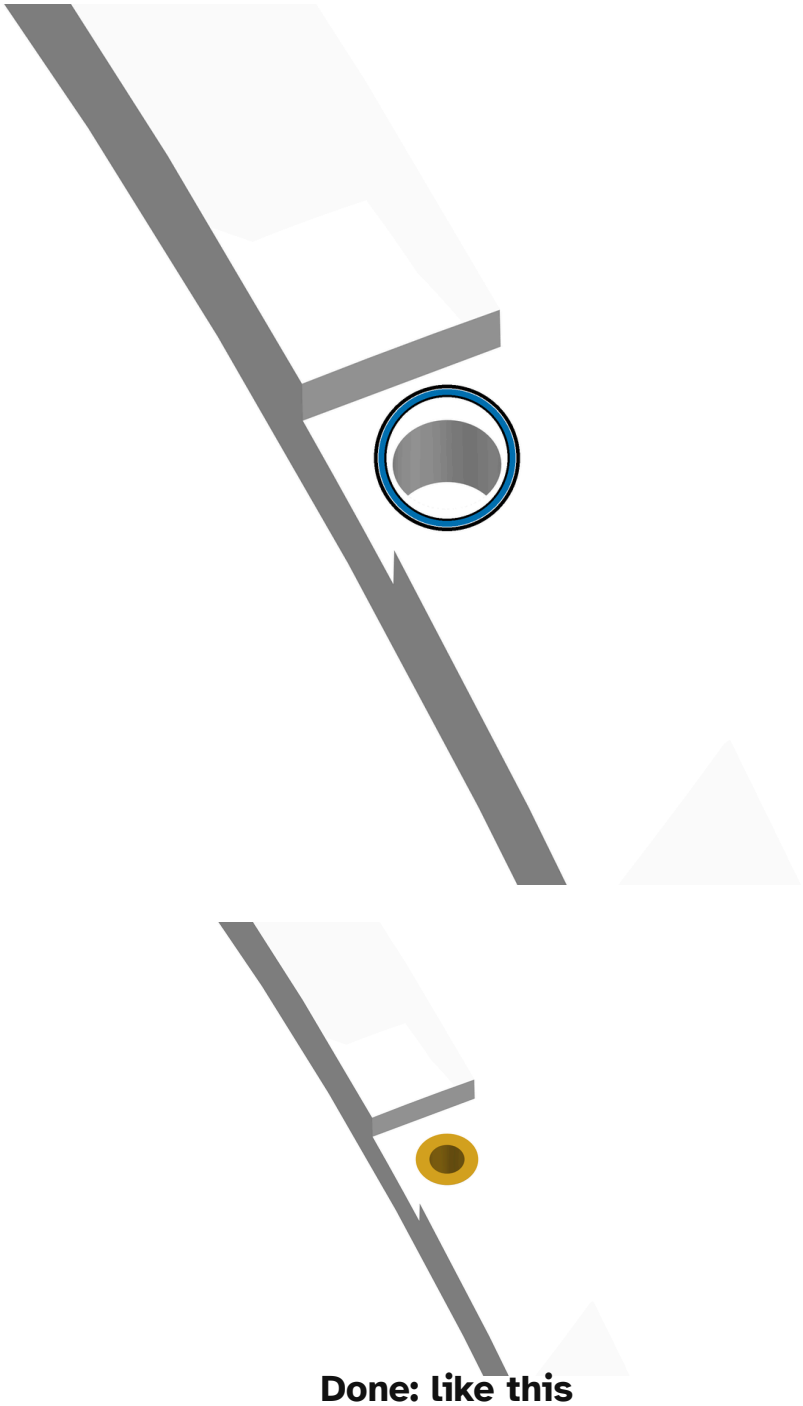
Done: like this

- Later a screw goes in here: sphere arcs.

Diagonal arc · LOWER, all 6

The middle, where the arcs cross

- **M3 SHORT 4 mm** ×1 · on each part: 6 parts = 6 inserts



Done: like this

- Later a screw goes in here: upper diagonal arc.
- Only the 6 LOWER arcs. The 6 upper arcs have no insert.

Good to know



No force

If it does not go in, wait. Heat, not muscle.



Wrong one?

Heat it again and pull it out with the iron.